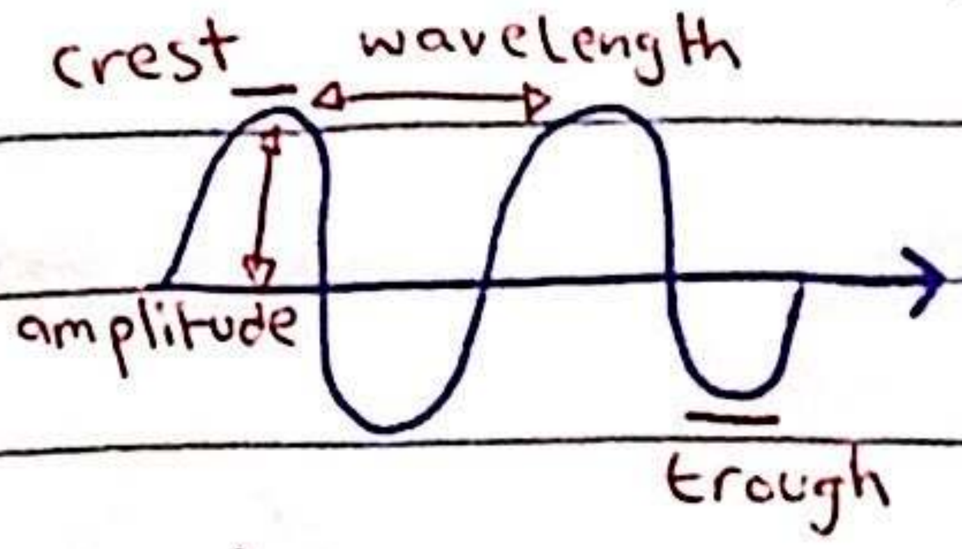
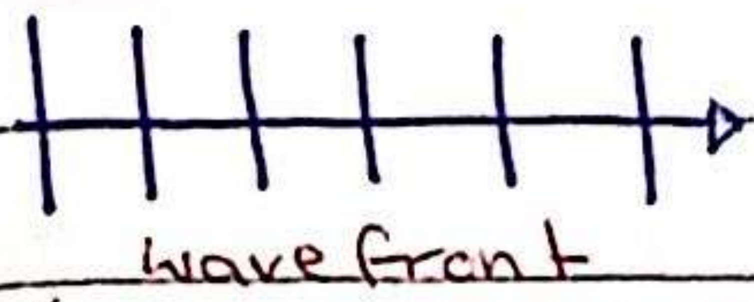
Block 3 18/12/22

Waves

- transfer of energy w/o any transfer of matter



'period' is the time taken for 1 oscillation



(wave from above)

frequency :: number of oscillations per unit time
(pitch)

amplitude :: max. displacement of moleculer from mean position
(Loudness)

wave speed = $f \lambda$ ← (wavelength)

Types of waves

- mechanical

waves that need a medium to go through

- electromagnetic

waves that don't need a medium to go through

- transverse

waves where the molecules vibrate perpendicular to the wave motion

- Longitudinal

waves where the molecules vibrate parallel to the wave motion

Behaviour of waves

- reflection

change in direction of wave when it strikes on a surface

- refraction

change in speed of a wave when the medium changes

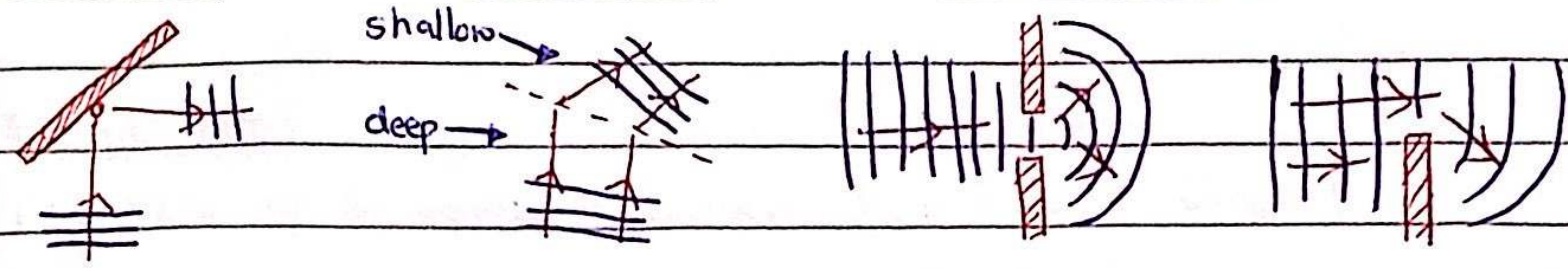
- diffraction

spreading of wave as it goes through a gap

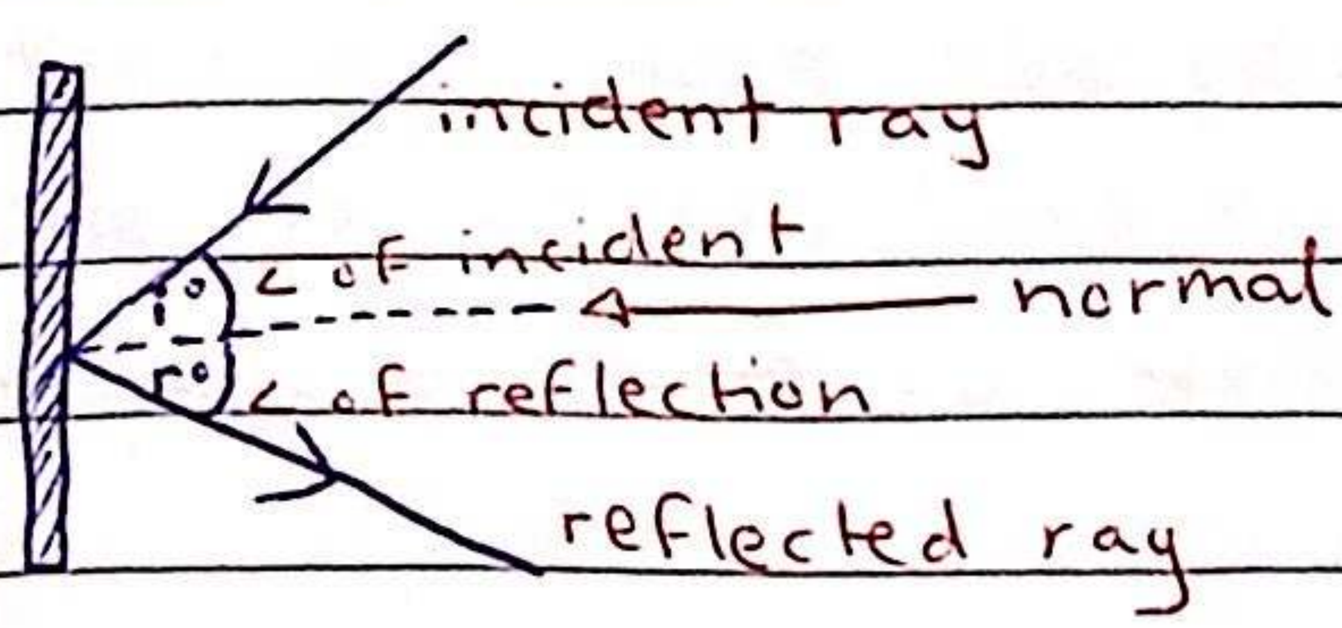
19/12/22

* in diffraction, if the gap is bigger, the wavelength will by lower and vice versa

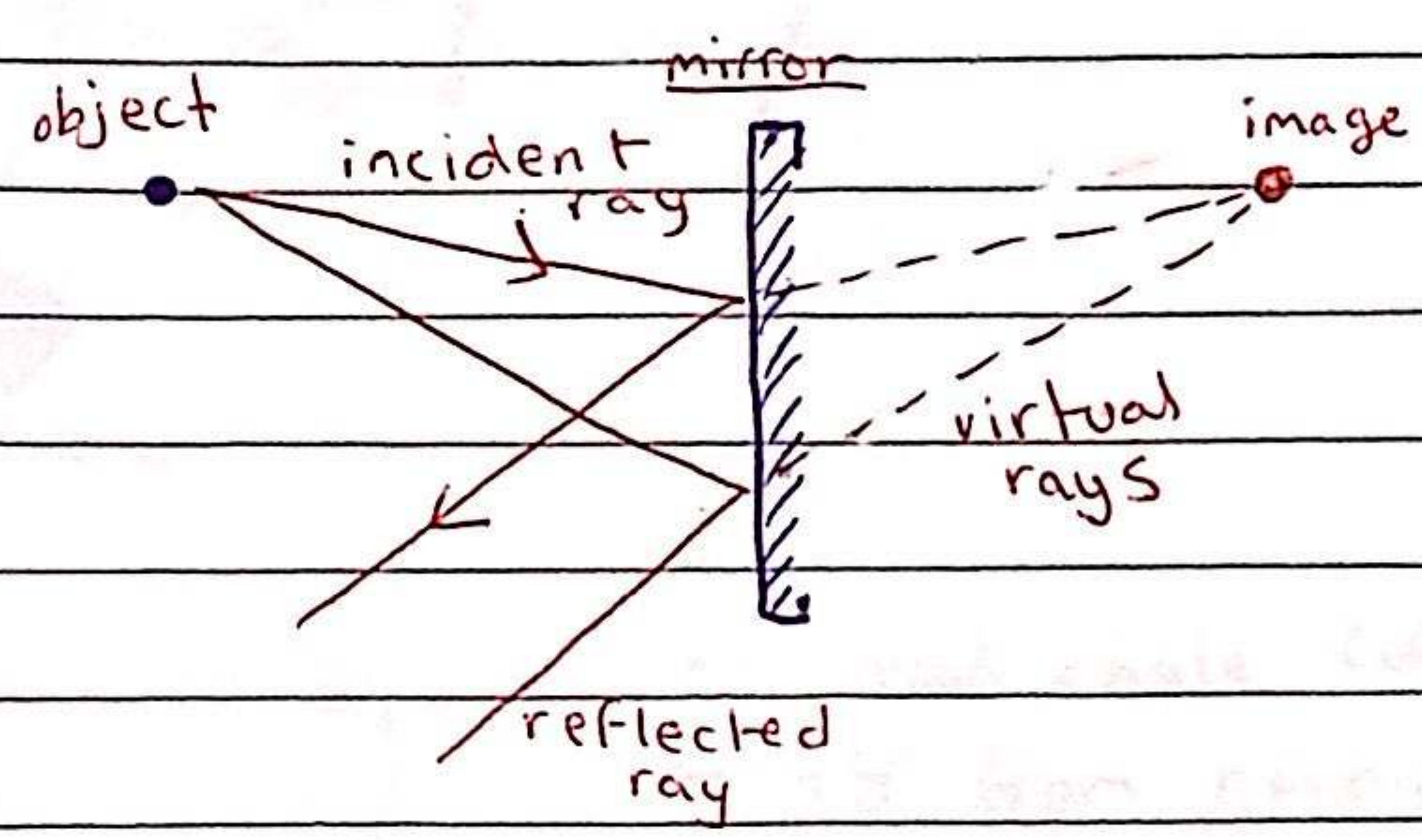
reflection refraction diffraction



reflection of light

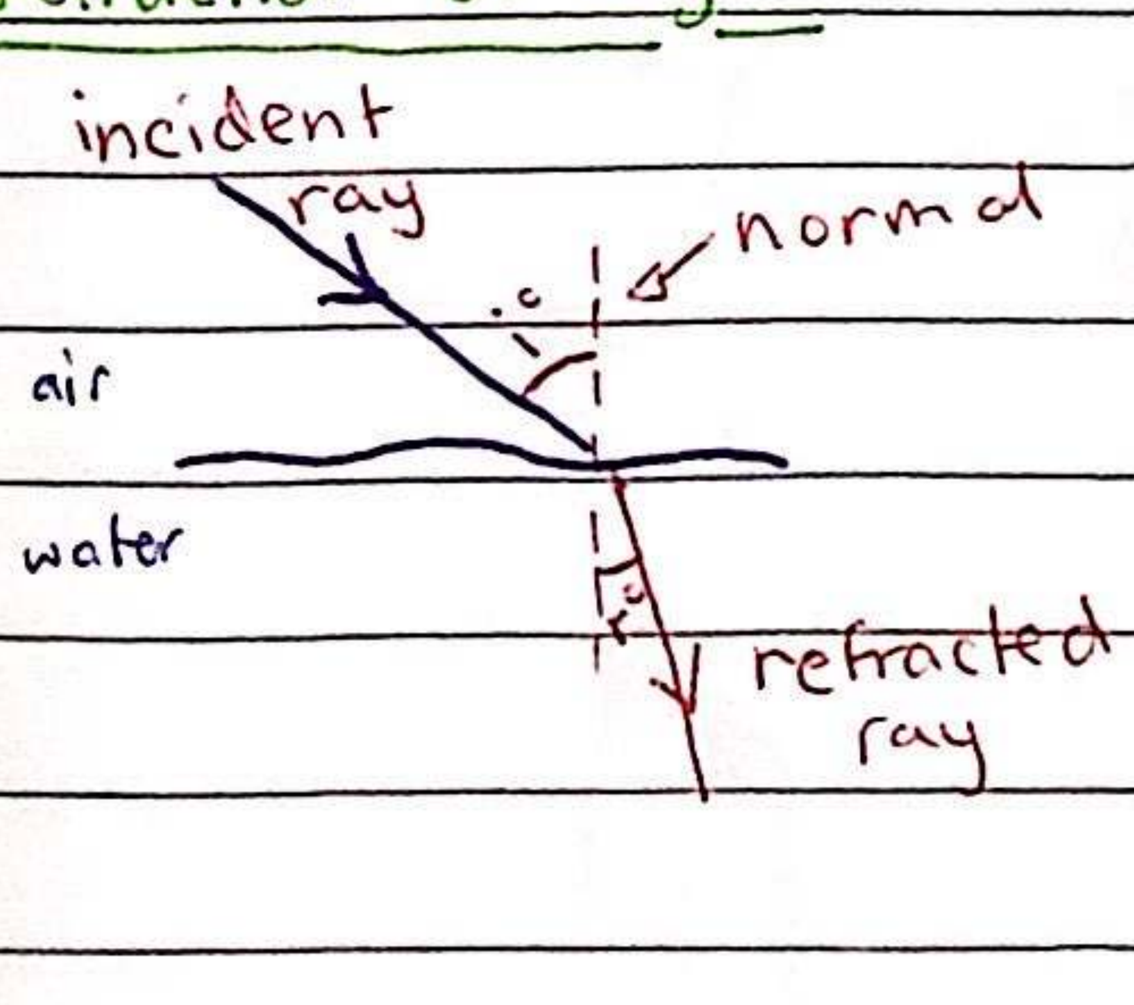


$$i^\circ = r^\circ$$



* the image formed in the mirror is known as a virtual image

refraction of light



* if the ray is perpendicular to the surface, there is no change in direction
* if the ray moves from less to more dense, it bends towards the normal
* if the ray moves from more to less dense, it bends away from the normal

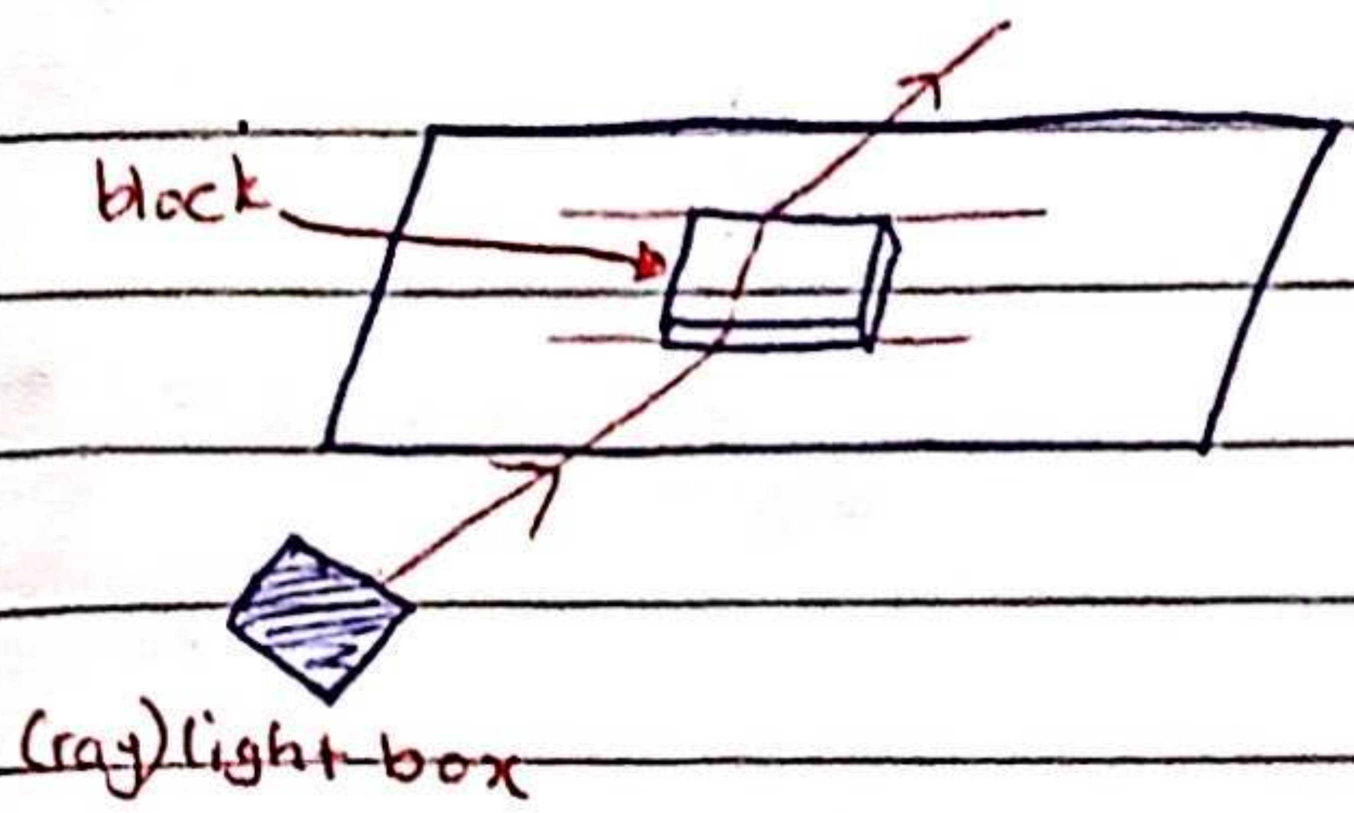
19/12/22

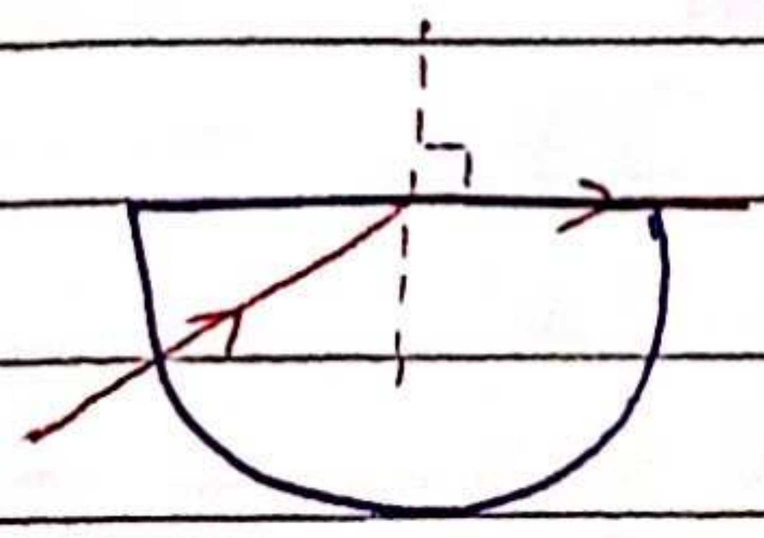
Investigating refraction

independant variable = shape of the block
dependant variable = direction of refraction
control variables = width/frequency/wavelength of light

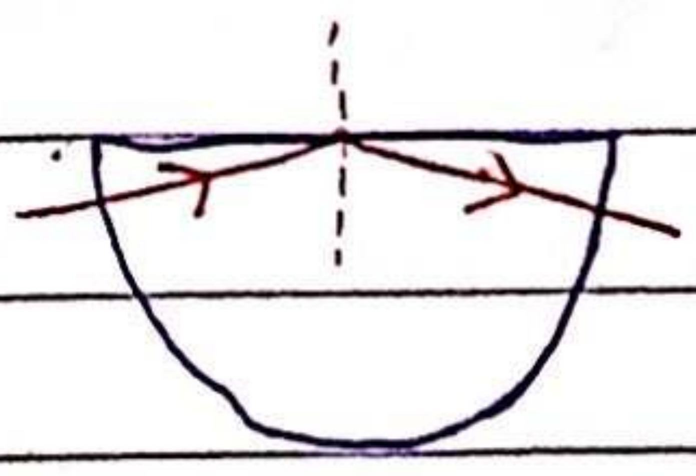
*** equipment**

- ray box → to provide a narrow beam of light
- protractor → to measure the angles of the light beam
- piece of paper → to mark lines on for angle measurement
- pencil → to mark the perpendicular/angle lines on the paper
- ruler → to draw lines on paper
- perspex block → to refract the light beam

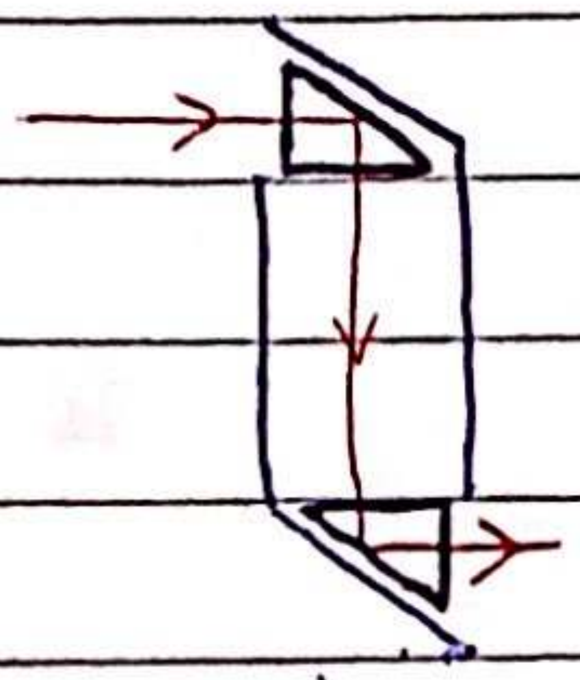




- critical angle (of incidence)
  ↳ 45° from normal



- total internal reflection
  ↳ $\angle i° = \angle r°$
  ↳ happens when all the light is reflected

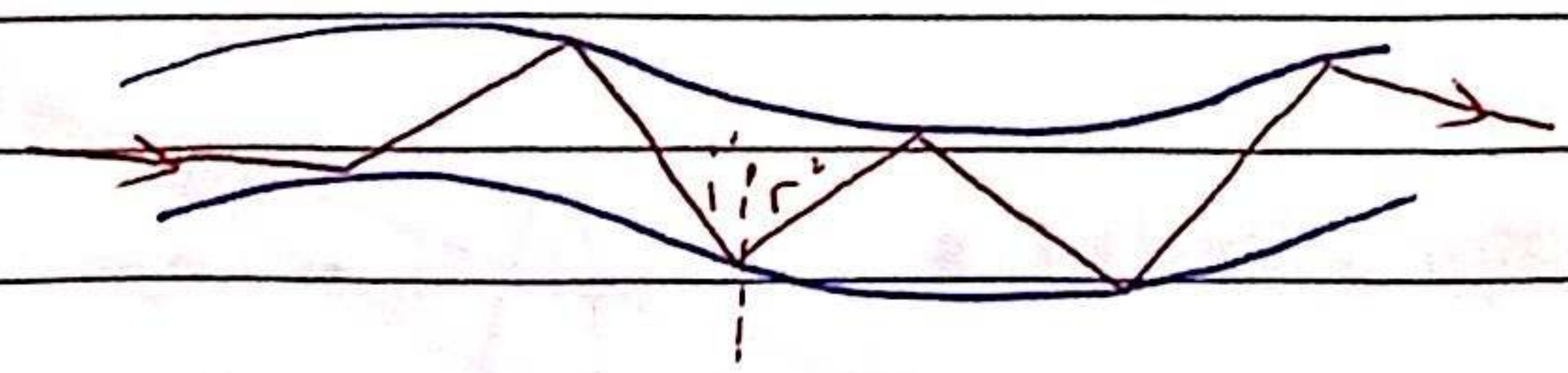


- the light totally internally reflects in both prisms

20/12/22

total internal reflection in along optical fibres ::

- light travelling down an optical fibre is to y internally reflected each time it hits the edge of the fibre

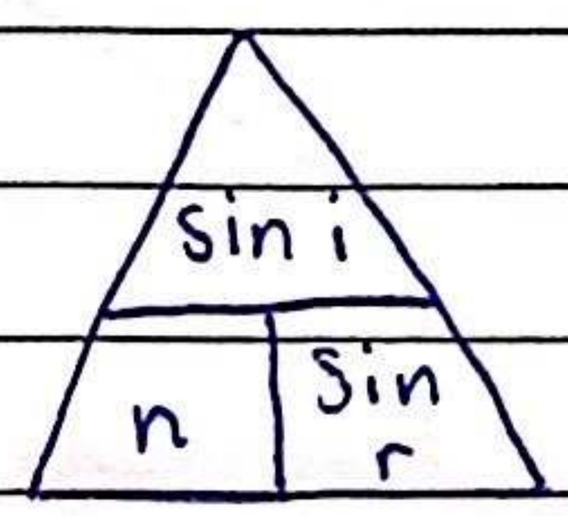


Refractive index

- the ratio of the speeds of waves in two different regions/mediums

→ $n = \frac{\sin i}{\sin r}$ ← ∠ of incidence / ← ∠ of refraction

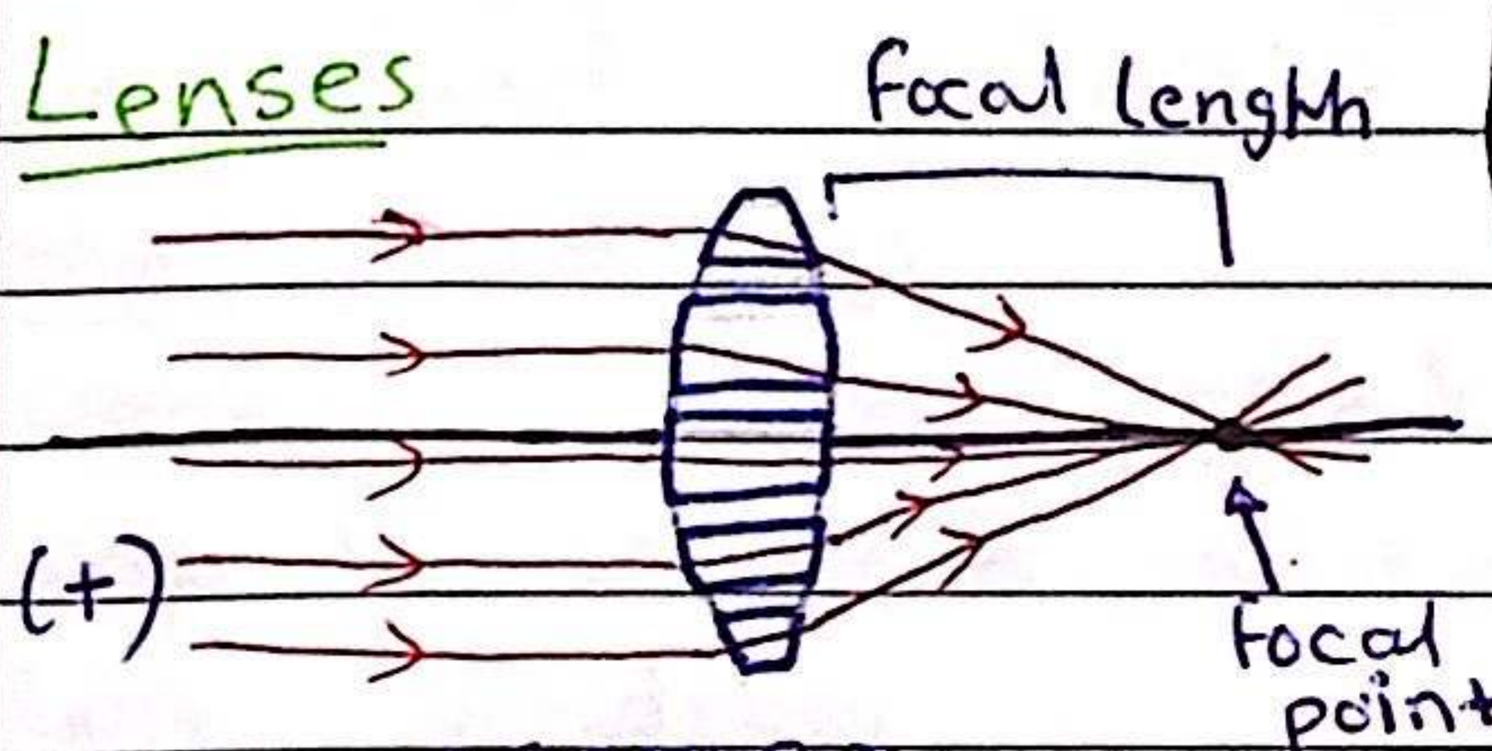
refractive index



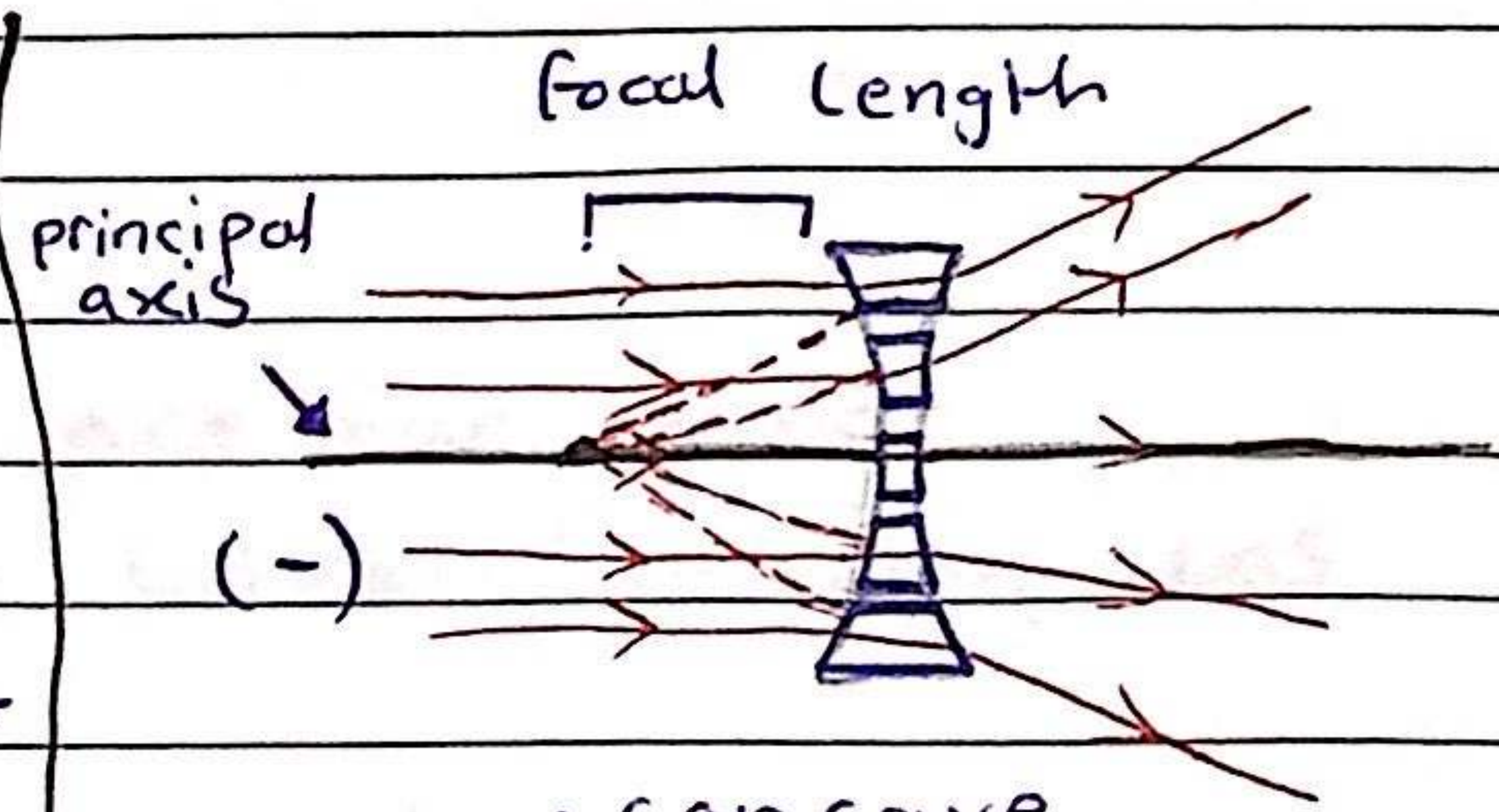
→ $n = \frac{1}{\sin c}$ ← critical angle

refractive index

Lenses

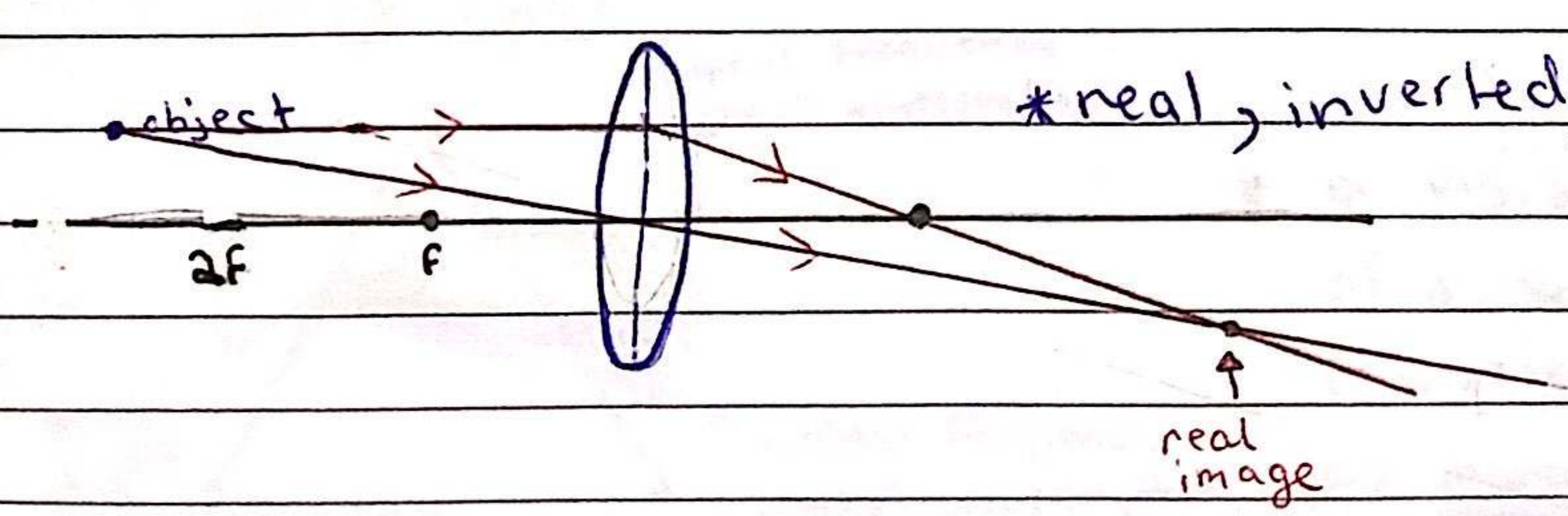


- convex
- converging



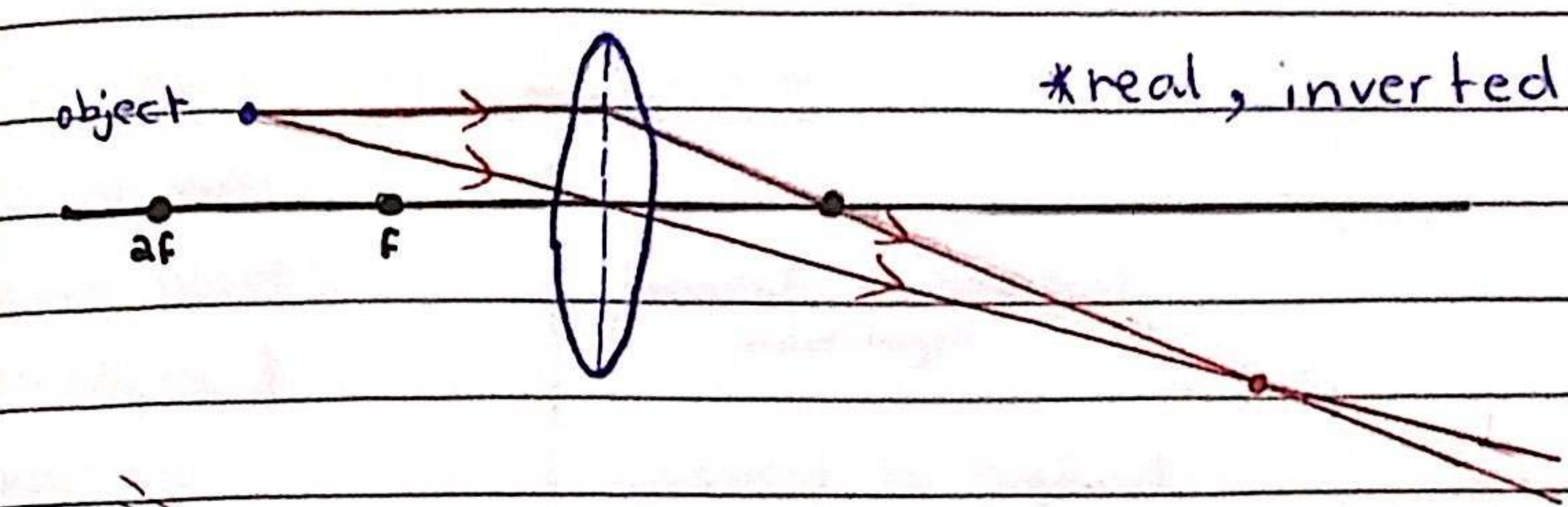
- concave
- diverging

Ray diagram

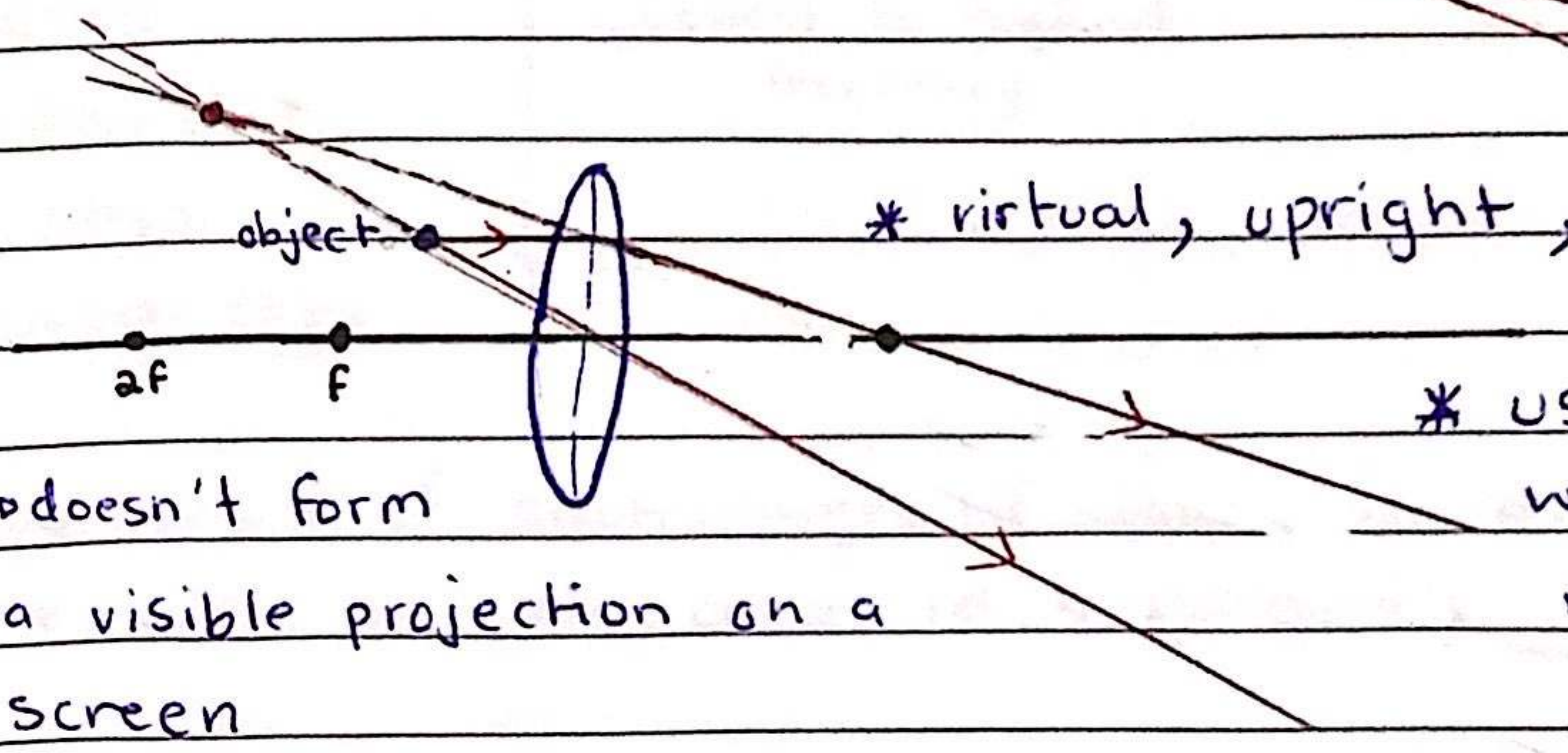


* real, inverted, diminished

20/12/22



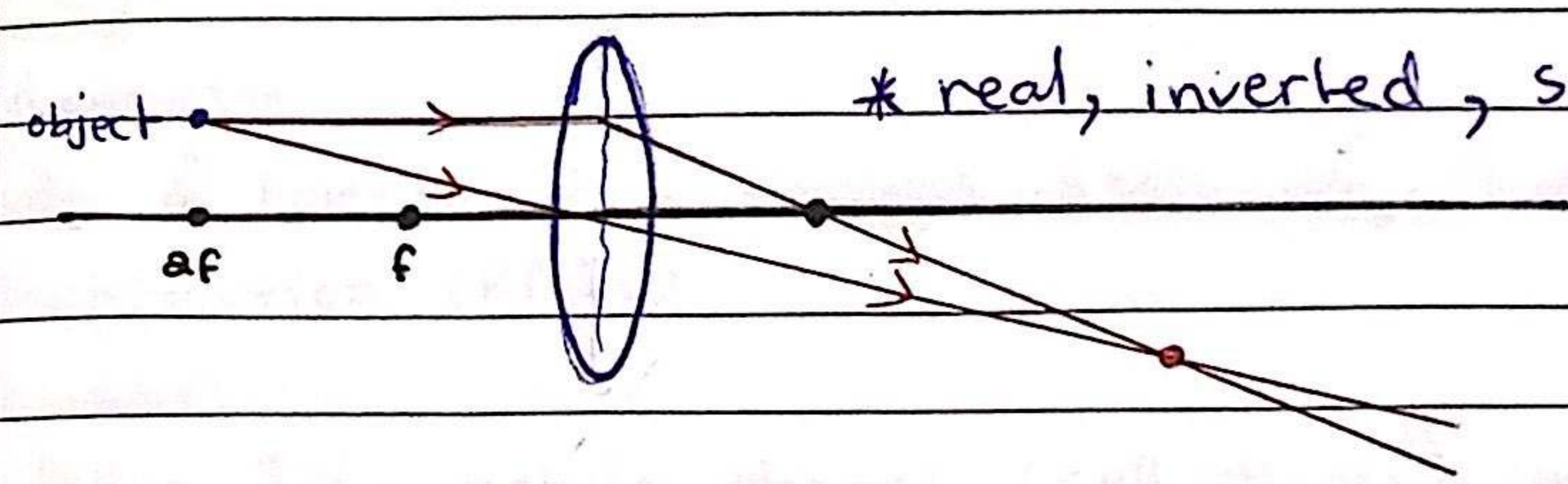
*real, inverted, enlarged



* virtual, upright, enlarged

* used in this way for a magnifying glass

→ doesn't form a visible projection on a screen



* real, inverted, same size

Short/Long sightedness:

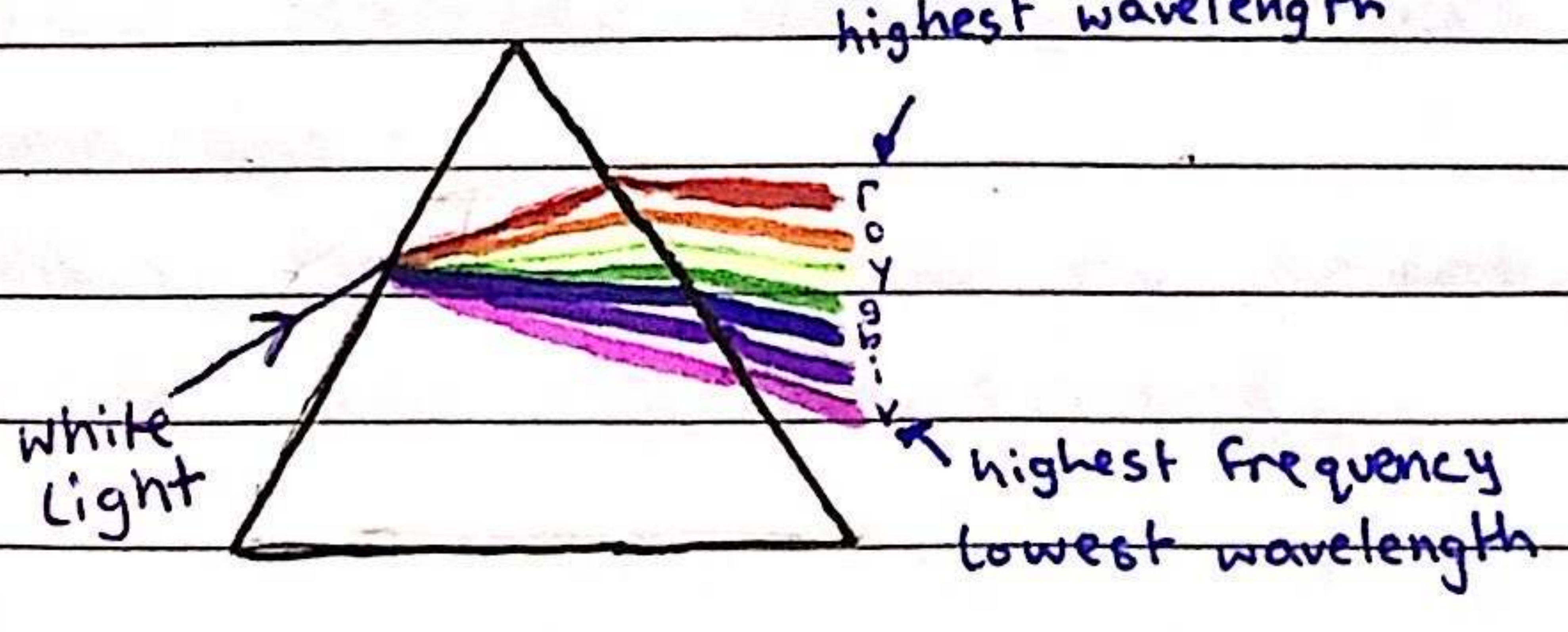
<u>Short sightedness</u>

- cannot see things that are far away
- can be correcte using concave/diverging lens

<u>Long sightedness</u>

- cannot see things that are close
- can be corrected using convex/converging lens

Dispersion of light



* a visible light of a single frequency or wavelength is known as <u>monochromatic</u>

20/12/22

Electromagnetic spectrum

- radio waves
- micro waves
- infrared
- visible
- ultraviolet
- x-rays
- gamma rays

- longest to shortest wavelength
- Lowest to highest frequency

* the speed of electromagnetic waves is approximately the same in air and in a vaccum; 3×10^8 m/s

Uses:-

radiowaves:-

radio & television transmissions, astronomy, radio frequency identification (RFID)

microwaves:-

Satellite TV, mobile phones (cell phones), microwave ovens

infrared:-

electric grills, short range communications such as nemote controllers for TV's, intruder alarms, thermal imaging, optical fibres

visible:-

vision, photography, illumination

ultraviolet:-

security marking, detecting fake bank notes, sterilising water

x-rays:-

medical scanning, security scanners

gamma rays:-

sterilising food, & medical equipment, detection of cancer and its treatment

20/12/22

Harmful effects :-

micro waves :-

internal heating of body cells

infrred :-

skin burns

ultraviolet :-

damage to surface cells & eyes, Leading to skin cancer and eye conditions

x-rays & gamma rays :-

mutation or damage to cells in the body

Artificial satelites :-

- communication is made mainly by microwaves
- Some satellite phones use low orbit artificial satellites while others use geostationary satellites (& direct broad satellite TV)

Electromagnetic radiation :-

- many important communication systems rely on this

cell phones / wireless internet :-

use microwave because micro waves can penetrate some walls and only require a short aerial fa transmission/reception

bluetooth :-

use radiowaves because radio waves can pass throgh walls but the signal is _____ on doing so

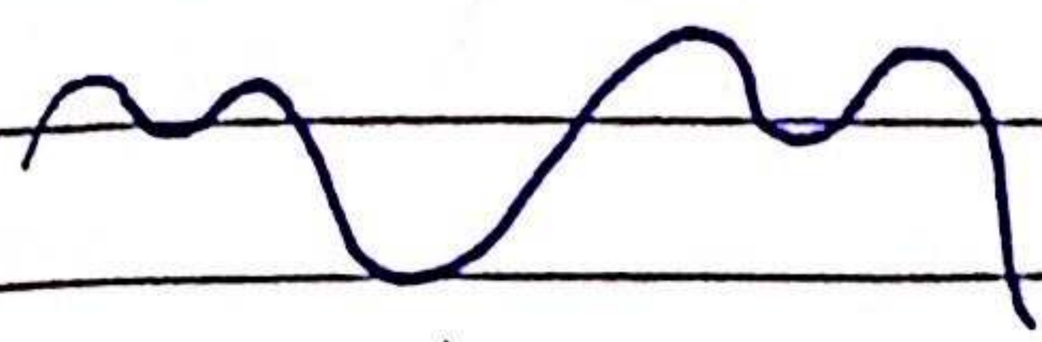
optical fibres :-

uses visible/infrared fore cable TV & high-speed broadband because glass is transparent to and some infrared light & visible light and short wave-Length infrared can carry high rates of data

20/12/22

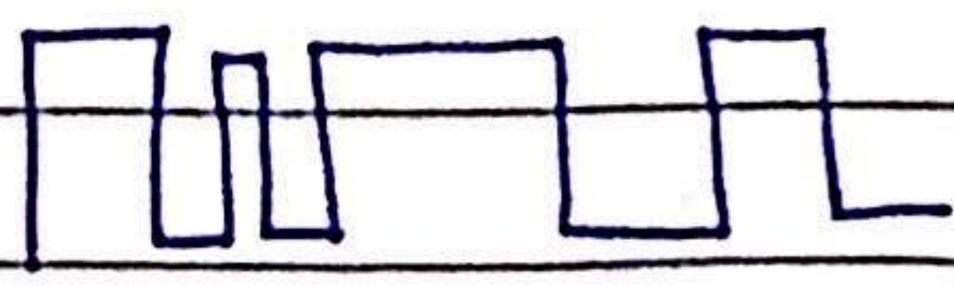
Digital /Analogue

- analogue signals can vary continuously, from any lowest value to any highest value (smoothly)
- digital signals are discrete, the only take particular values such as '1's & 0's', 'high & low', 'on & off'



analogue

e.g vinyl record, speedometer



digital

e.g CD, computer lightswitch

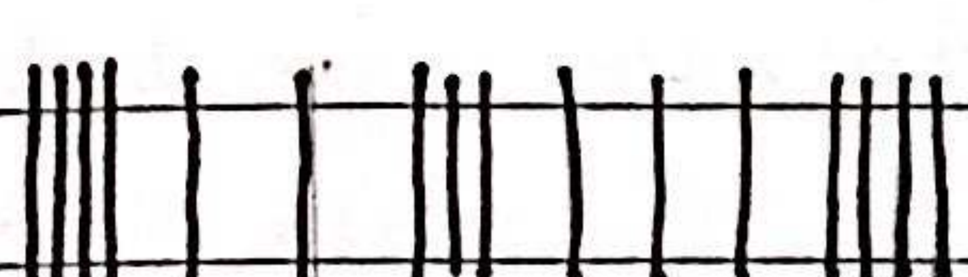
* sound can be transmitted in as a digital or analogue signal

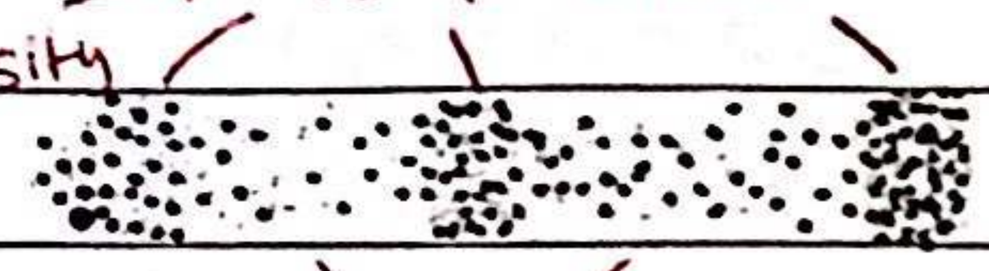
Benefits of digital:

+ increased rate of transmission of data
+ increased range due to accurate signal regeneration

Sound

- sound waves are produced by vibrating sources





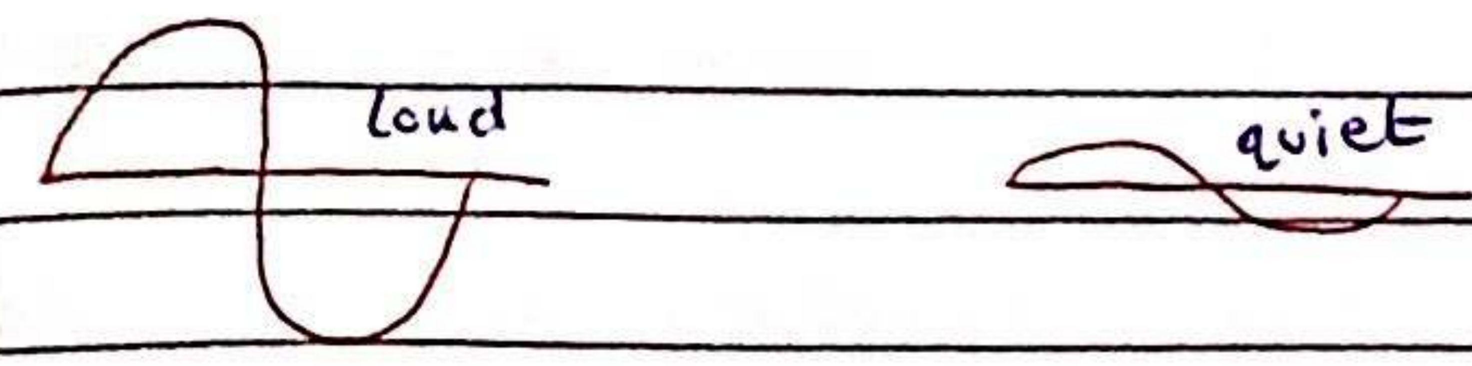
→ longitudinal wave

* the audible range of frequencies to humans is 20Hz - 20000Hz
* sound needs a medium to travel through
* speed of sound in air is 330 - 350 m/s
* speed of sound : solid > liquid > gases

* speed of sound = d/t

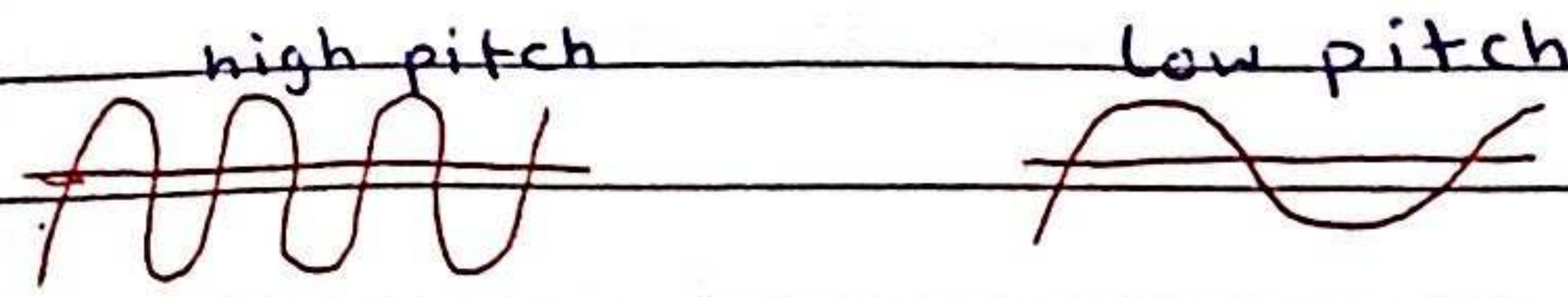
amplitude

- how loud a sound is



frequency

- the pitch of a sound



echo:-

- reflection of sound waves

ultrasound

- a sound with a frequency higher than 20kHz
- used in non-destructive testing of materials, medical scanning of soft tissue & sonar including calculation of depth or distance from time and wave speed.